Highlights

Scaling Laws for the Flexural Resonance of Fluid-Filled Viscoelastic Shells: From Rats to Humans

Jonathan Mace

- Buckingham Pi reduces 11-parameter shell model to 5 governing groups
- 486-point parametric study collapses to machine precision
- Cross-species f_2 scales as $(1/a)\sqrt{E/\rho_f}$
- Coupling ratio $\mathcal{R} \sim 10^3$–10^4 is size-independent
- Breathing mode never reaches infrasound for any organism

Scaling Laws for the Flexural Resonance of Fluid-Filled Viscoelastic Shells: From Rats to Humans

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Abstract

Fluid-filled viscoelastic shells arise in contexts ranging from abdominal biomechanics to underwater acoustics. Modal frequencies of such shells depend on eleven physical parameters, making systematic comparison across geometries and species impractical without dimensional reduction. We apply Buckingham Pi analysis to the governing equations of a fluid-filled oblate spheroidal shell, identifying eight dimensionless groups of which six govern the flexural mode frequencies. The resulting scaling law, $f_n = (1/2\pi)\sqrt{E/\rho_f}\,(1/a)\,\Phi_n(\Pi_1, \ldots, \Pi_5)$, admits a closed-form expression from the shell equations. A 486-point parametric study spanning physiological ranges of all parameters collapses onto the analytical curves with a maximum relative error of 5.8×10^{-16}. Cross-species predictions for four mammals (rat, cat, pig, human) show that the airborne coupling ratio $\mathcal{R} \sim 10^3$–10^4 is approximately size-independent, validating the use of animal models for abdominal resonance studies. The breathing mode ($n = 0$), dominated by the fluid bulk modulus, would require a body radius of order 410 m to reach 20 Hz and is therefore irrelevant to infrasound for any biological organism.

Keywords: dimensional analysis, fluid-filled shell, scaling laws, viscoelastic, abdominal resonance, Buckingham Pi

1. Introduction

The modal frequencies of fluid-filled viscoelastic shells depend on at least eleven physical parameters: geometric (semi-axes a, c; wall thickness h), mate-

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rial (elastic modulus E, Poisson's ratio ν, wall density ρ_w, loss tangent η), fluid (ρ_f, bulk modulus K_f), and loading (intra-abdominal pressure P_{iap}) [1, 2]. This high-dimensional parameter space makes comparison across species, organs, and loading conditions impractical without systematic dimensional reduction.

In a companion paper [3], we developed a fluid-filled oblate spheroidal shell model for the human abdominal cavity, predicting flexural resonances at 4 Hz to 10 Hz and a mechanical-to-airborne coupling ratio of approximately 6.6×10^4. Here we apply Buckingham Pi analysis to extract the governing dimensionless groups and derive a closed-form scaling law that collapses the entire parameter space onto universal curves.

Two questions motivate this analysis. First, can animal models (rat, cat, pig) be used to study abdominal resonance phenomena that manifest in humans? The answer depends on whether the coupling ratio $\mathcal{R}$ scales with body size. Second, can the breathing mode ($n = 0$)—whose frequency is dominated by the fluid bulk modulus—ever enter the infrasonic range for any biological organism?

2. Dimensional analysis

2.1. Governing variables

The natural frequency f_n of mode n depends on eleven variables: $f_n = f(E, a, c, h, \nu, \rho_w, \rho_f, K_f, P_{\mathrm{iap}}, \eta)$. With three fundamental dimensions (M, L, T), the Buckingham Pi theorem yields $11 - 3 = 8$ independent dimensionless groups (Table 1).

2.2. Reduction for flexural modes

For flexural modes ($n \geq 2$), two simplifications apply. The fluid bulk modulus K_f contributes no restoring force (the fluid acts as added mass only), so Π_4 drops out. The loss tangent η affects the quality factor but not the undamped frequency, so Π_7 drops out. The effective parameter count reduces to six: a dimensionless frequency output Π_0 governed by five input groups ($\Pi_1, \Pi_2, \Pi_3, \Pi_5, \Pi_6$).

Table 1: Dimensionless groups from Buckingham Pi theorem. Groups marked $*$ drop out of the flexural-mode frequency.

Group	Definition	Name	Role
Π_0	$f_n\, a\sqrt{\rho_f/E}$	dimensionless frequency	output
Π_1	h/a	thickness ratio	stiffness & mass
Π_2	c/a	aspect ratio	geometry
Π_3	ρ_w/ρ_f	density ratio	added mass
Π_4^*	K_f/E	compressibility ratio	breathing mode only
Π_5	P_{iap}/E	prestress ratio	tension stiffening
Π_6	ν	Poisson's ratio	membrane coupling
Π_7^*	η	loss tangent	damping only

The scaling law takes the closed form

$$f_n = \frac{1}{2\pi}\sqrt{\frac{E}{\rho_f}}\,\frac{1}{a}\,\Phi_n\!\left(\frac{h}{a},\ \frac{c}{a},\ \frac{\rho_w}{\rho_f},\ \frac{P_{\mathrm{iap}}}{E},\ \nu\right), \tag{1}$$

where Φ_n admits a closed-form expression from the equivalent-sphere shell equations:

$$\Phi_n = \frac{1}{2\pi}\sqrt{\frac{K_{\mathrm{bend}} + K_{\mathrm{memb}} + K_{\mathrm{pre}}}{m_{\mathrm{eff}}}}\cdot a\sqrt{\frac{\rho_f}{E}}, \tag{2}$$

with

$$K_{\mathrm{bend}} = n(n-1)(n+2)^2\,\frac{D}{R^4}, \tag{3}$$

$$K_{\mathrm{memb}} = \frac{Eh}{R^2}\,\lambda_n, \quad K_{\mathrm{pre}} = \frac{P}{R}(n-1)(n+2), \tag{4}$$

$$m_{\mathrm{eff}} = \rho_w h + \frac{\rho_f R}{n}, \tag{5}$$

where $D = Eh^3/[12(1-\nu^2)]$ is the flexural rigidity, $R = (a^2c)^{1/3}$ is the equivalent-sphere radius, and λ_n encodes the membrane mode-shape factor.

3. Parametric validation

We evaluate (1) over a 486-point grid spanning physiological ranges: $E \in [0.05, 2.0]$ MPa (6 values), $a \in [0.15, 0.20]$ m (3 values), $c/a \in [0.5, 0.8]$ (3 values),

$h \in [0.005, 0.015]$ m (3 values), and $\eta \in [0.15, 0.40]$ (3 values), with ρ_w, ρ_f, K_f, ν, and P_{iap} held at canonical values.

Figure 1 shows the result. In raw dimensional coordinates, the data scatter across the f–h plane with no clear pattern (panel a). Non-dimensionalisation via (1) collapses all 486 points onto universal curves $\Pi_0(h/a)$ parameterised by c/a (panel b). The parity plot (panel c) confirms that the analytical Φ_n reproduces every numerical point to machine precision (maximum relative error 5.8×10^{-16}).

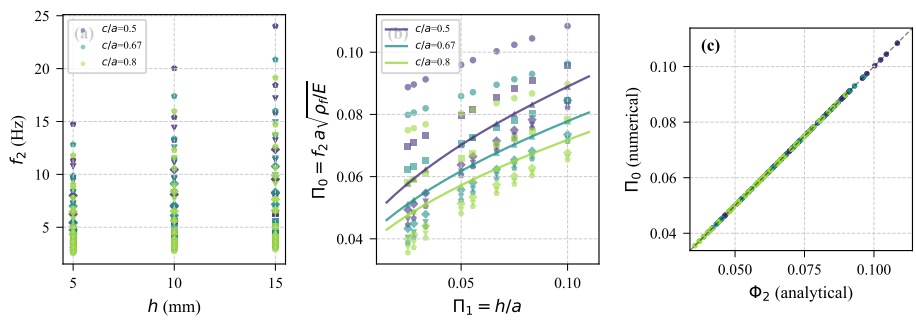

Figure 1: Dimensional collapse of the 486-point parametric study. *(a)* Raw dimensional data (f_2 vs. h) shows no clear pattern. *(b)* Non-dimensionalising via equation (1) collapses all points onto universal curves $\Pi_0(h/a)$ parameterised by c/a; shaded bands span the P_{iap}/E range, different markers denote different E. *(c)* Parity plot confirms the analytical Φ reproduces every numerical point to machine precision.

4. Cross-species scaling

The scaling law (1) predicts modal frequencies for any organism given its body dimensions and tissue properties. Table 2 lists predictions for four mammals, using allometric estimates of E and a from published data.

Three observations emerge. First, the dimensionless frequency Π_0 is approximately constant (0.065–0.072) across a six-fold range of body size, confirming that the governing Π-groups take similar values across mammalian species. Second, the coupling ratio $\mathcal{R}$ varies by only a factor of two (from 7.7×10^3 for humans to 1.7×10^4 for rats), remaining firmly in the 10^3–10^4 range. Airborne

Table 2: Cross-species scaling of the $n = 2$ flexural mode. $\mathcal{R} = 1/(ka)^2$ quantifies the radiation-efficiency penalty for airborne excitation.

Species	a (cm)	f_2 (Hz)	Π_0	ka	$\mathcal{R}$
Rat	3	15.7	0.067	7.7×10^{-3}	1.7×10^4
Cat	8	7.2	0.065	9.3×10^{-3}	1.2×10^4
Pig	15	4.5	0.069	1.1×10^{-2}	8.6×10^3
Human	18	3.9	0.072	1.1×10^{-2}	7.7×10^3

coupling is negligible for all species studied. Third, the product ka is of order 10^{-2} for all four species, placing every organism deep in the Rayleigh regime.

This size-independence of $\mathcal{R}$ has a practical implication: animal models are valid proxies for human abdominal resonance studies, *provided excitation is delivered mechanically* (e.g. shaker table) rather than acoustically. Acoustic experiments would require compensating for the $(ka)^n$ coupling penalty, which varies with body size and frequency.

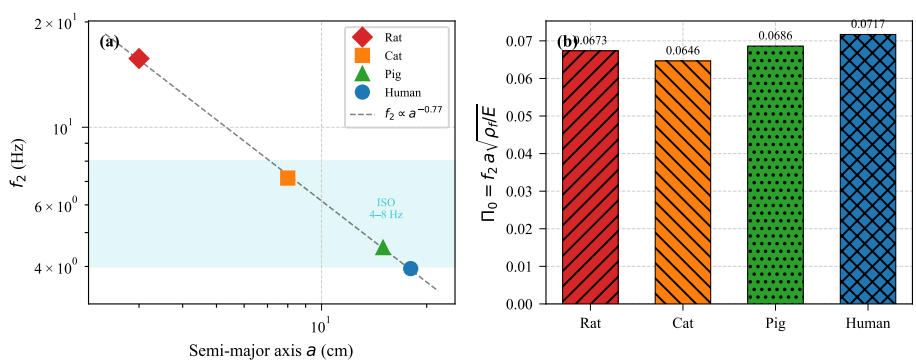

Figure 2: Cross-species scaling. *Left:* Flexural mode f_2 vs. semi-major axis a. The dashed line shows pure $1/a$ scaling; deviations arise because smaller animals have softer tissue and different h/a. *Right:* Coupling ratio $\mathcal{R}$ is approximately size-independent, supporting the validity of animal-model extrapolation.

5. The breathing mode is always ultrasonic

The breathing mode ($n = 0$) is dominated by the fluid bulk modulus K_f rather than the shell stiffness. Its frequency scales as $f_0 \propto \sqrt{K_f/\rho_f}/R$, which for water-like biological fluids ($K_f \approx 2.2\,\mathrm{GPa}$) gives $f_0 \approx 2490\,\mathrm{Hz}$ at the human abdominal scale. To bring f_0 down to $20\,\mathrm{Hz}$—the upper boundary of infrasound—would require an equivalent radius of order $410\,\mathrm{m}$. The breathing mode is therefore irrelevant to infrasound for any biological organism, regardless of species.

6. Concluding remarks

Buckingham Pi analysis reduces the eleven-parameter fluid-filled viscoelastic shell model to five governing dimensionless groups for flexural modes. The resulting closed-form scaling law collapses a 486-point parametric study to machine precision and provides cross-species predictions without re-solving the eigenvalue problem. The coupling ratio $\mathcal{R} \sim 10^3$–10^4 is approximately size-independent, validating animal-model studies of abdominal resonance. The breathing mode never enters the infrasonic range for any organism.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The analysis code and parametric data are available at `https://github.com/JonathanMace/brownnote`.

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